MACHINE WHITE PAPERS

UNISIG Deep-Hole Drilling Machine — Deep Hole

Precision gundrilling and BTA drilling for long, straight, tight-tolerance holes in oilfield and aerospace parts. Long-bore capability — high L/D ratios.

Deep Hole · gundrilling / BTA deep-hole drilling machine

Deep, straight, concentric holes are one of the hardest features to produce on any CNC platform without a purpose-built machine. Most job shops don't have one. Downhole tool bodies, valve stems with long bores, hydraulic cylinders, and any part that needs a straight bore longer than a few diameters routinely gets outsourced — or worse, made poorly on an inadequate setup. UNISIG deep-hole drilling is a genuine capability moat.

What Deep Hole is for

Deep Hole does the one thing that is nearly impossible to fake: long, straight, on-size holes at high length-to-diameter ratios. Gundrilling and BTA drilling are separate disciplines from turning and milling, with their own tooling, their own coolant pressure requirements, and their own failure mode — drift. A hole that wanders half a degree over 30 inches is scrap, and you usually don't find out until it's finished. Shops without dedicated deep-hole equipment either subcontract this work or attempt it with a twist drill and hope.

Pick this machine when

- The bore's length-to-diameter ratio is beyond what a conventional drill can hold straight.
- Straightness and positional accuracy of the bore are specified, not assumed.
- Downhole tool bodies, long hydraulic cylinders, manifolds with deep internal passages, gun-drilled shafting.
- Surface finish inside the bore matters — sealing bores, hydraulic bores, bores that will be honed afterwards.

When it's the wrong machine

Short holes. A 3:1 hole is a drilling operation on whatever machine is already holding the part, and routing it here adds a setup for no benefit. Blind pockets and non-round internal features aren't deep-hole work either, whatever their depth.

From the floor

High-pressure through-tool coolant is not optional and it is not primarily about cooling — it is chip evacuation. In a deep bore the chip has one way out, and a packed chip is what breaks the drill and scores the bore on the way. Coolant pressure and flow get set for the specific bore diameter and depth before the hole is started, because there is no recovering a deep hole partway through.

Pick your industry

UNISIG deep-hole drilling for Oil & Gas

Full spec sheet, typical oil & gas parts, alloys, tolerance expectations, documentation approach.

UNISIG deep-hole drilling for Aerospace

Full spec sheet, typical aerospace parts, alloys, tolerance expectations, documentation approach.

UNISIG deep-hole drilling for Military & Defense

Full spec sheet, typical defense parts, alloys, tolerance expectations, documentation approach.

UNISIG deep-hole drilling for Industrial & General Manufacturing

Full spec sheet, typical industrial parts, alloys, tolerance expectations, documentation approach.

Have a project?